

The well-balanced reconstruction and the hydrostatic condition

Abstract

The Well balanced: True option of EXHALE is the reconstruction of Käppeli & Mishra (2016): every cell carries its own hydrostatic equilibrium, the reconstruction and the Riemann solver act on the departure from it, and the momentum source is the face-pressure difference of that same equilibrium, so that at a discrete hydrostatic equilibrium the flux difference and the source cancel in the algebra and not in floating point. This memo states the connection term by term, maps it onto the code, and records why it is the one change that lets the LHS 1140 b base layer (Mach 2.7×10^{-7}) reach a certified stationary state: the ordinary reconstruction's truncation error in the face pressure enters the HLLC contact speed as a spurious velocity one to three decades above the physical one, and no solver can remove a residual the scheme itself manufactures. Every number is measured on the tree of 2026-09-13 unless marked otherwise.

1. Background: well-balanced schemes

A scheme is called well balanced for a given steady state when its discrete operator vanishes exactly on the discrete counterpart of that state, so that the state is preserved to rounding and small departures from it are resolved at the scheme's accuracy instead of being buried under the truncation error of two large cancelling terms. The idea entered Godunov-type methods with LeVeque (1998), who noted that a high-resolution method applied to a balance law with a source term produces spurious waves near a quasi-steady state unless the flux gradient and the source are discretized so as to cancel, and proposed a quasi-steady wave-propagation algorithm that splits the cell so that the source balances the flux difference inside it. Botta et al. (2004) treated the case that matters here, nearly hydrostatic flows of an atmosphere, where the pressure gradient and gravity are each many orders larger than their difference: they showed that the standard finite-volume discretization loses its accuracy in that regime and built a well-balanced method around a local hydrostatic reconstruction of the pressure.

Käppeli & Mishra (2014) gave a well-balanced finite-volume scheme for the Euler equations with a general gravitational potential by defining, inside each cell, a local equilibrium at the cell's own specific entropy and integrating the mechanical balance $h + \phi = \text{const}$ (their eq. 2.7); density and pressure are both extrapolated to the faces from that isentropic equilibrium and the momentum source is the face-pressure difference of the same equilibrium (their eqs. 2.14 and 2.26), which their theorem 1 shows to be a second-order consistent discretization of $-\rho d\phi/dx$. Its limitation is that the equilibrium preserved exactly is the isentropic one. Käppeli & Mishra (2016) removed that limitation: the equilibrium is again the mechanical balance, but the subcell integral is taken with a constant density inside the cell, so no thermodynamic variable other than the cell's own pressure and density enters, and “the reconstruction preserves discrete hydrostatic equilibria with arbitrary temperature or entropy stratification” (their section 2.1.1). Density and velocity keep the standard reconstruction, the pressure is reconstructed as a departure from the local equilibrium, and the only requirement on the numerical flux is that it resolve a stationary contact discontinuity exactly. That is the variant EXHALE implements (decision 23 of 2026-09-10). Provenance: the design memo of 2026-09-10 read the authors' ETH SAM research reports (2015-40 and 2013-05, the submitted texts); the published articles (references/Kappeli_2016A&A_587_A94.pdf, references/Kappeli_2014JCP_259_199.pdf) were obtained

on 2026-09-13 and every equation number quoted here and in the design memo was checked against them: eqs. (10), (11), (13) to (18) and (24) to (27) of the 2016 article, eqs. (2.7), (2.9) to (2.14), (2.23), (2.25), (2.26) and theorem 1 of the 2014 article carry the same numbers and the same content as in the reports, and the sentence quoted from section 2.1.1 is verbatim. The two papers were read in full; the other references of this section were consulted for their scope and are cited for it.

Related constructions exist. Chandrashekar & Klingenberg (2015) give a second-order well-balanced finite-volume scheme for the Euler equations with gravity that preserves isothermal and polytropic hydrostatic states exactly with any consistent numerical flux, by reconstructing the pressure and density as departures from those particular equilibria. Edelmann et al. (2021) compare well-balanced treatments of gravity in astrophysical (stellar) hydrodynamics, where the same near-hydrostatic cancellation limits the accessible Mach number, and show on stellar models that without such a treatment the spurious velocities exceed the physical convective ones at low Mach number. The two ingredients of the EXHALE scheme that the well-balanced reconstruction leans on are older: the HLLC approximate Riemann solver of Toro et al. (1994), which restores the contact wave the HLL solver averages out and therefore resolves a stationary contact exactly, and the WENO reconstruction in the efficient form of Jiang & Shu (1996), whose third-order variant is the production reconstruction of EXHALE (Reconstruction scheme: WENO3).

2. The hydrostatic condition in a finite-volume scheme

The stationary momentum equation of the wind is

$$\frac{d}{dr}(\rho v^2) + \frac{dp}{dr} = -\rho \frac{d\phi}{dr} - \frac{2\rho v^2}{r}, \quad (1)$$

and in the base layer of LHS 1140 b, where the flow is at Mach 2.7×10^{-7} , it reduces to the hydrostatic balance $dp/dr = -\rho d\phi/dr$ to one part in 10^{-13} . The two terms of that balance are each 4.4×10^5 times the wind's momentum flux in the same layer (*Update_EXHALE* stage 1, section 143); the wind is their difference.

A finite-volume scheme forms the two terms in different places. The pressure enters through the numerical flux at the faces, $F_{j+1/2}$, built from face states that a reconstruction (PLM or WENO3) extrapolates from the cell averages; gravity enters as a cell-centered source, $-\rho_j \phi'(r_j)$. The update of the momentum row is

$$R_{2,j} = \frac{A_{j+1/2} F_{2,j+1/2} - A_{j-1/2} F_{2,j-1/2}}{V_j} + \rho_j \phi'(r_j), \quad (2)$$

and at a hydrostatic column the two contributions cancel only up to the truncation error of the reconstruction, of order $(\Delta r/H)^p$ times the $O(1)$ terms, p being the order of the reconstruction and H the pressure scale height. That residual is not small against the wind: on the LHS 1140 b column the WENO3 face pressure differs from the cell pressure by 5.7×10^{-5} relative at $r = 1.001 R_p$ (measured on the archived state; 7.3×10^{-6} after eight 1-2-1 smoothings of the column and 1.2×10^{-7} on a ninth-order polynomial fit, converging at order 2.97 to 3.00 under refinement), while the stationary velocity is 2.7×10^{-7} of the sound speed.

Where that error does its damage is the HLLC flux. Its contact speed is

$$S^* = \frac{p_R - p_L + \rho_L v_L (S_L - v_L) - \rho_R v_R (S_R - v_R)}{\rho_L (S_L - v_L) - \rho_R (S_R - v_R)} \equiv S_{\text{vel}}^* + S_{\text{pres}}^*, \quad S_{\text{pres}}^* = \frac{p_R - p_L}{\phi_L - \phi_R}, \quad (3)$$

and the face energy flux is $F_3 = S^*(E^* + p^*)$. In the base cells of LHS 1140 b, S_{vel}^*/c_s is 3.4×10^{-7} to 1.1×10^{-6} , the Mach number of the gas, while S_{pres}^*/c_s is 5×10^{-6} to 9×10^{-4} : the pressure-jump part is one to three decades larger. The energy flux divergence of the base cells is then the divergence of a spurious velocity, and the energy row reads $O(1)$ of its own largest term in every cell below $1.05 R_p$ (measured -4.6×10^4 times the net heating at cell 2, $+2.7 \times 10^3$ at cell 5, -52 at cell 20, $+1.7$ at cell 41, where the velocity part alone is $+3.4, +1.1, +1.2, +1.5$, i.e. within a factor two of the stationary balance). The seed does not matter: three seeds whose base velocity differed by a factor 10^4 gave the same energy maximum to every printed digit. A Newton solver cannot remove a residual that the scheme manufactures from the pressure field alone.

3. The well-balanced reconstruction

3.1 Each cell carries its own hydrostatic equilibrium

Within cell j , hold the density at the cell value ρ_j and integrate the mechanical balance from the cell's own pressure:

$$p_{\text{eq},j}(r) = p_j - \rho_j [\phi(r) - \phi(r_j)]. \quad (4)$$

This is eq. (16) of Käppeli & Mishra (2016). It states nothing about the temperature or the entropy, which is why it preserves an arbitrary stratification, and it is second-order consistent with $dp/dr = -\rho\phi'$ in the cell width. Its two face values are the arrays `wb_P_up(j)` and `wb_P_dn(j)` of `hydrostatic_equilibrium_of_each_cell` (`src/modules/states/Reconstruction.f90`):

$$P_{\text{up}}(j) = p_j - \rho_j [\phi(r_{j+1/2}) - \phi(r_j)], \quad (5)$$

$$P_{\text{dn}}(j) = p_j + \rho_j [\phi(r_j) - \phi(r_{j-1/2})]. \quad (6)$$

The potential is the analytic function of r the code already evaluates at faces and centers (`Gphi_i`, `Gphi_c`), so the staggered interpolation of the paper's eq. (15) is not needed.

3.2 The reconstruction carries the departure

The perturbation data of cell j is the neighbor's pressure measured against cell j 's equilibrium continued *through the face* to the neighbor's center, with cell j 's density up to the face and the neighbor's beyond it:

$$d_+(j) = (p_{j+1} - p_j) + \rho_j \phi_{\text{up}}(j) + \rho_{j+1} \phi_{\text{dn}}(j+1), \quad (7)$$

$$d_-(j) = (p_{j-1} - p_j) - \rho_j \phi_{\text{dn}}(j) - \rho_{j-1} \phi_{\text{up}}(j-1), \quad (8)$$

with $\phi_{\text{up}}(j) = \phi(r_{j+1/2}) - \phi(r_j)$ and $\phi_{\text{dn}}(j) = \phi(r_j) - \phi(r_{j-1/2})$, and the value at the cell's own center identically zero. The limited slope (PLM) or the WENO3 stencil is applied to the triple $(d_-, 0, d_+)$ exactly as it is applied to (p_{j-1}, p_j, p_{j+1}) otherwise, with the same limiter, the same smoothness indicators and the same volume-share coefficients, and returns two face departures. The face pressures handed to the Riemann solver are $P_{\text{up}}(j) + q_L(j)$ and $P_{\text{dn}}(j+1) + q_R(j)$; density and velocity keep their ordinary reconstruction.

The discrete hydrostatic equilibrium this preserves is eq. (18) of the paper,

$$\frac{p_{j+1} - p_j}{\Delta r} = -\frac{\rho_j + \rho_{j+1}}{2} \frac{\phi_{j+1} - \phi_j}{\Delta r}, \quad (9)$$

which is exactly the statement that the two neighboring extrapolations agree at the shared face, $P_{\text{up}}(j) = P_{\text{dn}}(j+1)$: the face mismatch

$$\delta p_{\text{eq}}(j+1/2) = P_{\text{dn}}(j+1) - P_{\text{up}}(j) = d_+(j) \quad (10)$$

(`wb_dp_eq` in `well_balanced_face_departures`) vanishes on it. On the discrete equilibrium the perturbation data is therefore identically zero, the reconstruction returns the equilibrium itself, and the face is a stationary contact discontinuity: equal pressure on both sides and zero velocity. The only requirement on the flux is that it resolves one exactly, $F([\rho_L, 0, p], [\rho_R, 0, p]) = [0, p, 0]$, which HLLC and Roe do and LLF does not (the reason the key warns under `Numerical flux: LLF`). Continuing cell j 's own density over the whole gap to the neighboring center instead of switching to the neighbor's beyond the face leaves data of size $(\rho_j - \rho_{j+1})$ times the potential difference across half a cell, which does not vanish on eq. (9): measured, that form leaves 2.5×10^{-5} of the momentum weight on the discrete column at $N = 250$ where the form above leaves 3.9×10^{-14} .

3.3 The source is the same equilibrium

The momentum source is no longer the cell-centered $-\rho_j \phi'(r_j)$ but the face-pressure difference of the cell's own equilibrium,

$$S_{2,j} = \frac{A_{j+1/2} P_{\text{up}}(j) - A_{j-1/2} P_{\text{dn}}(j)}{V_j} \quad \text{measured against the face pressures,} \quad (11)$$

which is a second-order consistent discretization of $-\rho d\phi/dr$ (Käppeli & Mishra, 2014, theorem 1); in the code `Source` returns $S(2, j) = 0$ and the momentum row is assembled from the face pressures measured against the equilibrium. At the discrete equilibrium the flux difference and the source are then the same expression and cancel in the algebra; away from it the departure carries the ordinary second-order accuracy of the reconstruction, so the wind region is discretized as before.

3.4 What is and is not preserved

The equilibrium of eq. (4) holds the density constant inside a cell. What the scheme preserves exactly is therefore the *discrete* equilibrium (9), not the continuous hydrostatic solution $p(r)$ with its own $\rho(r)$; the two agree to second order in the cell width. The consequence is that the base layer of a well-balanced solution converges with the grid at the same order as before, and the grid convergence of the LHS 1140 b solution has not yet been measured (open item L5c). Nothing thermal is assumed: the temperature stratification of the base layer is whatever the energy equation returns.

4. What it did on LHS 1140 b

The measurements are those of item L5c on the reference case (GJ 1132 proxy spectrum, $\text{He/H} = 1.6261$, $K_{zz} = 10^9 \text{ cm}^2 \text{ s}^{-1}$, scalar base at 226 K and $1 \mu\text{bar}$), the archived 2026-08-30 solution mapped onto the current grid as the seed, the partitioned stationary route (`Restart intent: stationary`, `EXHALE_PTC_DTAU0=1.0`), 16 threads.

Table 1: The hydrodynamic rows of outer pass 1 (cellwise maximum of $|R|$ over the row's own largest term) and the outcome of the stationary solve, same seed, input and binary.

option	hydro	mass	momentum	energy	s	outcome
none	info= 2	1.08	3.2×10^{-2}	0.63	1325	pass refused
Low-Mach damping: $2.0\text{e-}2$	info= 2	1.32	1.4×10^{-2}	1.24	1232	pass refused
Well balanced: True	info= 0	2.6×10^{-10}	1.8×10^{-14}	4.6×10^{-9}	85	CERTIFIED at pass 8

With the key on, the LHS 1140 b atomic wind with binary H/He element diffusion is certified at outer pass 8 in about ten minutes (mass 4.5×10^{-10} , momentum 2.3×10^{-13} , energy 5.1×10^{-9} , gated element row 8.8×10^{-6} of 10^{-5} , face mass-flux spread 1.5×10^{-11}). Three seeds whose base velocities differed by a factor 10^4 (the mapped archived state, the same with the stationary velocity imposed below $1.27 R_p$, and a Bernoulli-consistent temperature) reach one solution: ρ to 3.4×10^{-10} , v to 2.5×10^{-8} , $\dot{M}$ to seven digits. The element relaxation, which refused every pass without the key (element_step_out_of_bounds after 24 to 26 steps, the helium fraction driven to unity by a hydrodynamic state whose mass flux fell 27 percent across the wind), reaches its fixed point in 32 steps in every pass with it. Low-Mach damping does nothing to the energy row because it is a fourth difference of the *velocity* and does not touch the pressure truncation error. On the HD 209458 b element-diffusion reload the key changes nothing that matters: certified at the same outer pass 12, $\dot{M}$ moved by 1.3×10^{-4} , profiles beyond $2 R_p$ by 2 to 3×10^{-4} , the base cells by 12 to 14 percent. The solution itself differs from the archived one: $\log_{10} \dot{M} = 8.29$ against 7.80, the base cell at 405 K against 252 K, wind densities three to ten times higher. Part of that is the physics that moved since 2026-08-30 (a heating 3.4 times larger in the first cells after the photon-grid quadrature correction, the Jupiter radius, the spectrum below 13.6 eV), part the base condition, and part the fact that the archived state was never stationary below $1.1 R_p$ under the present residual (the old gates began at $1.2 R_p$). Which part is which is the comparison of item L4 of the plan and is not settled here.

5. Why the key is off by default

Turning it on changes the discretization of the pressure/gravity pair in every run, not only in the base layer: the two discretizations are both second-order consistent and neither is a subset of the other, so every golden of the regression matrix would move (md/input_schema.md K46). The decision of 2026-09-10 (decision 23) made it an option, stated in EXHALE_setup.out, carried as the option token wellbal in every state file, and left off. For LHS 1140 b it is required: without it there is no certifiable state of the wind on the current code, with any seed and by any route (plan items L1 to L5c). The decision of 2026-09-13 is to run every LHS 1140 b model with the key on and to say so in each case's REPRODUCE.md. Whether it should become the default for every planet is a separate question, to be decided on the regression matrix and the Model A comparison, not here.

References

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